MODULE 33 · DOMAIN 4 - RESILIENCE, MIGRATION AND CONTINUITY

Rehosting with AWS Transform MGN

Replicate servers continuously into AWS, test without touching production, cut over in minutes, and know when relocating the whole VMware estate fits better.

DOMAIN 4 of 5 EXAM WEIGHT 15% MODULE 33 of 52

admin@sthithapragna.com

Rehosting is not a copy. It is **continuous replication**, so the cutover is a short switch rather than a long outage.

The MGN lifecycle

01
Replicate
Agent or vCenter appliance streams changes

02
Stage
Low-cost replication servers and EBS

03
Test
Launch test instances, validate

04
Cut over
Stop source, final sync, launch

05
Finalise
Archive source, stop replication

ON THE EXAM

Test as many times as needed; replication keeps running between tests.

Agent or agentless replication?

REPLICATION AGENT

On each server

- Physical, virtual or other clouds
- Any supported hypervisor
- Agent installed in the guest OS
- The default and most general

AGENTLESS

Through vCenter

- vCenter client appliance on-premises
- VMware VMs only
- Nothing installed in the guests
- Snapshot-based replication

THE TIEBREAKER

No software allowed inside VMware guests: agentless replication through the vCenter appliance.

Network and staging design

- **Replication traffic** over Direct Connect or VPN, using private IPs.
- **Staging subnet**: private, sized for replication servers.
- **Interface endpoints** for MGN, EC2 and S3 keep control traffic private.
- **Bandwidth throttling** protects the production network link.
- **Encrypted in transit**; replicated data stays in your VPC.

> ON THE EXAM
>
> Replication must not cross the internet: private IP over Direct Connect, plus interface endpoints.

Launching safely

- **Launch templates** set instance type, subnet, security groups, licensing.
- **Right-size** from discovery data, not the source hardware.
- **Isolated test subnet** so test instances cannot reach production systems.
- **Domain controllers**: launch first, or keep them in the source.
- **Applications and waves** group servers to test and cut over together.

> ON THE EXAM
>
> Test instances that can reach production directories or databases can cause real damage.

Post-launch actions

ACTION	PURPOSE
Install SSM agent	Manage the new instance immediately
Install CloudWatch agent	Metrics and logs from day one
Upgrade Windows OS	Clone and upgrade an out-of-support version
Convert licences	Move to AWS-provided licensing
Custom SSM documents	Your own hardening and configuration

ON THE EXAM

Standard changes on every migrated server, without logging in to each one: post-launch actions.

Rehost to EC2, or relocate with EVS?

REHOST WITH MGN

VMs become EC2 instances

- Per-server replication and cutover
- EC2 operations, pricing, scaling
- Leaves VMware licensing behind
- Path to later modernisation

RELOCATE WITH AMAZON EVS

Run VMware in your VPC

- VMware Cloud Foundation on AWS metal
- Same vSphere tools and runbooks
- Bring your own VCF licence
- You operate the VCF stack

THE TIEBREAKER

Exit fast, change nothing in operations, keep the VCF licences: relocate to Amazon EVS.

AWS Transform for rehosting

- **Agentic workflow**: discovery, waves, network, landing zone, rehost.
- **Network conversion**: target VPCs generated from source segments as CloudFormation.
- **Rehosting runs on MGN**; switch to the MGN console at any time.
- **Humans approve** plans and changes before they run.

> ON THE EXAM
>
> A large VMware exit wanting automated planning and network build: AWS Transform orchestrating MGN.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Minimal downtime, many servers	MGN continuous replication
No agents in VMware guests	Agentless replication via vCenter
Replication must stay private	Private IP over DX or VPN, endpoints
Testing must not affect production	Isolated test subnet
Keep vSphere operations, BYO VCF licence	Amazon EVS
Same fix-ups on every migrated server	Post-launch actions

ON THE EXAM

Application Migration Service in an older question is AWS Transform MGN in a newer one.

> Meridian Freight must move 300 Linux and Windows servers, a mix of physical and VMware, to EC2. Each application can be down for at most 30 minutes during cutover.
>
> Teams want to test each application in AWS several times before cutover.

What should the solutions architect use?

A AWS Transform MGN with replication agents, test launches, then cutover

B VM Import/Export of each server's disk image

C AWS DataSync to copy each server's files

D Rebuild every server from new AMIs and reinstall applications

◆ PAUSE AND DECIDE

What should the solutions architect use?

A MGN with agents

Continuous replication; repeatable tests; minutes of cutover downtime.

B VM Import/Export

One-off image copy, long downtime, VMware only.

C DataSync

Copies files, not bootable servers.

D Rebuild

That is a replatform effort, and slow.

THE SIGNAL

Many servers, physical and virtual, minimal downtime, repeated tests: MGN with agents.

> In another Meridian Freight data centre, all 600 servers are VMware virtual machines. The security team forbids installing any additional software inside guest operating systems.
>
> The servers must be rehosted to EC2 with minimal downtime.

How should replication be configured?

A Install the MGN replication agent on each VM

B Use VM Import/Export

C Use AWS Elastic Disaster Recovery agents

D Deploy the MGN vCenter client appliance and use agentless replication

◆ PAUSE AND DECIDE

How should replication be configured?

A Agent per VM

Installs software in the guests, which is forbidden.

B VM Import/Export

No continuous replication, so long downtime.

C DRS agents

Still an agent inside each guest.

D Agentless via vCenter

Replicates from vCenter snapshots; nothing in the guests.

THE SIGNAL

All VMware, nothing in the guests: agentless replication.

> Meridian Freight's financial regulator forbids migration data crossing the public internet. The company has a 10 Gbps Direct Connect connection to its landing zone.
>
> Replication for 200 servers is about to start.

What should the solutions architect configure?

A Default replication settings, since data is encrypted in transit

B Private IP replication over Direct Connect, a private staging subnet, and interface endpoints for MGN and related services

C Ship disks on AWS Snowball devices

D A Site-to-Site VPN over the internet

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Defaults

Encrypted, but may use public paths.

B Private over Direct Connect

Data and control traffic stay on private paths.

C Snowball

Closed to new customers, and no continuous replication.

D VPN over internet

Encrypted, but still crosses the public internet.

THE SIGNAL

Not over the internet: private IPs on Direct Connect, plus interface endpoints.

> During Meridian Freight's first wave, a test instance of the order service started processing real orders from the on-premises message broker, because it could reach the corporate network.
>
> The team must keep testing before cutover without this happening again.

What should the solutions architect change?

A Stop testing and cut over directly

B Launch test instances in the production subnet

C Set the launch template for tests to an isolated subnet with no route to on-premises production systems

D Shut down the on-premises broker during tests

◆ PAUSE AND DECIDE

What should the solutions architect change?

A Skip testing

Removes the safety the test stage provides.

B Production subnet

Makes the problem worse.

C Isolated test subnet

Tests run on real data without touching production.

D Stop the broker

Causes a production outage for every test.

THE SIGNAL

Test copies are full copies: isolate them from anything real.

> Meridian Freight must leave a co-location facility within four months. Its 2,000 VMware VMs are run by a team with mature vSphere and NSX runbooks, and the company has recently bought VMware Cloud Foundation licences.
>
> Leadership wants no change to VMs or operating procedures during the move.

What should the solutions architect recommend?

A Relocate to Amazon Elastic VMware Service in the company's VPC, using its VCF licences

B Rehost all 2,000 VMs to EC2 with MGN

C Refactor the applications to containers

D Retain on-premises in a new facility

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Amazon EVS

Same VMware stack and runbooks, in AWS, with owned licences.

B Rehost to EC2

Changes the operating model, which leadership ruled out for now.

C Refactor

Far too slow for four months.

D New facility

Not a cloud migration at all.

THE SIGNAL

No change to VMs or operations, VCF licences owned, tight deadline: relocate with EVS.

> Every server Meridian Freight migrates needs the Systems Manager and CloudWatch agents, a hardening script, and, for 40 servers, an upgrade from an unsupported Windows Server version.
>
> The team wants this done automatically as part of each launch.

What should the solutions architect configure?

A. A runbook for engineers to log in to each instance after cutover

B. User data scripts written separately for each server

C. EC2 Image Builder pipelines for every source server

D. MGN post-launch actions: predefined agent installs and Windows upgrade, plus a custom SSM document for hardening

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Manual runbook

Hundreds of logins; slow and inconsistent.

B Per-server user data

Custom scripting for each server.

C Image Builder

Builds new images; rehosted servers are replicated, not built.

D Post-launch actions

Applied automatically to every test and cutover launch.

THE SIGNAL

Standard changes on every rehosted server: post-launch actions.

What to carry out of this module

- **AWS Transform MGN** is Application Migration Service, renamed June 2026.
- **Continuous replication**, repeatable tests, minutes of cutover.
- **Agent anywhere**; **agentless** for VMware with no guest software.
- **Private replication**, isolated tests, post-launch actions.
- **Relocate with EVS** when VMware operations must not change.

ON THE EXAM

Replicate, stage, test, cut over, finalise.